
```
function [x, k] = metoda_jacobi(A, b, tol, max_iter)
    % A - matricea sistemului
    % b - vectorul termenilor liberi
    % tol - toleranta acceptata
    % max_iter - numarul maxim de iteratii (pentru a evita bucle infinite)
    % x - solutia aproximata
    % k - numarul de iteratii efectuate

    n = length(b);
    x = zeros(n, 1);
    x_nou = zeros(n, 1);

    D = diag(diag(A));
    M = D;
    N = D - A;

    for k = 1:max_iter

        x_nou = M \ (N * x + b);

        if norm(x_nou - x, inf) < tol
            x = x_nou;
            return;
        end

        x = x_nou;
    end

    warning('metoda Jacobi nu a convers ');
end

Not enough input arguments.

Error in metoda_jacobi (line 9)
    n = length(b);
        ^
```

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```
function [x, k, w_opt] = metoda_sor(A, b, w, tol, max_iter)
```

```
n = length(b);  
x = zeros(n, 1);  
x_nou = zeros(n, 1);
```

```
D = diag(diag(A));  
L = tril(A, -1) * -1; % partea strict inferioara  
U = triu(A, 1) * -1;  % partea strict superioara
```

```
% calculul lui w optim  
T_jacobi = D \ (L + U);  
rho = max(abs(eig(T_jacobi)));
```

```
w_opt = 2 / (1 + sqrt(1 - rho^2));
```

```
if isempty(w)  
    w = w_opt;  
end
```

```
M = (1/w)*D - L;  
N = ((1-w)/w)*D + U;
```

```
for k = 1:max_iter
```

```
    x_nou = M \ (N * x + b);
```

```
    if norm(x_nou - x, inf) < tol  
        x = x_nou;  
        return;  
    end
```

```
    x = x_nou;
```

```
end
```

```
warning('metoda SOR nu a convers');
```

```
end
```

```
Not enough input arguments.
```

```
Error in metoda_sor (line 4)
```

```
    n = length(b);  
        ^
```

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```

n = 10;

A = rand(n, n);
for i = 1:n
    % ne asiguram ca elementul de pe diagonala este mai mare decat
    % suma modulelor celorlalte elemente de pe linie.
    suma_linie = sum(abs(A(i, :))) - abs(A(i, i));
    A(i, i) = suma_linie + rand() + 1;
end

x_exact = (1:n)';

b = A * x_exact;

tol = 1e-6;
max_iter = 1000;

[x_jacobi, iter_jacobi] = metoda_jacobi(A, b, tol, max_iter);
eroare_jacobi = norm(x_jacobi - x_exact, inf);
fprintf('Jacobi:          Iteratii = %3d, Eroare = %e\n', iter_jacobi,
eroare_jacobi);

[x_gs, iter_gs, ~] = metoda_sor(A, b, 1, tol, max_iter);
eroare_gs = norm(x_gs - x_exact, inf);
fprintf('Gauss-Seidel: Iteratii = %3d, Eroare = %e\n', iter_gs, eroare_gs);

[x_sor, iter_sor, w_o] = metoda_sor(A, b, [], tol, max_iter);
eroare_sor = norm(x_sor - x_exact, inf);
fprintf('SOR (w=%.3f): Iteratii = %3d, Eroare = %e\n', w_o, iter_sor,
eroare_sor);

Jacobi:          Iteratii = 54, Eroare = 3.452521e-07
Gauss-Seidel: Iteratii = 12, Eroare = 6.223258e-08
SOR (w=1.191): Iteratii = 18, Eroare = 2.834781e-07

```

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